KEY TAKEAWAYS

1st ESPI Centre for Sustainability and Resilience Workshop

EUROPEAN COOPERATION IN SPACE SUSTAINABILITY

The yearly number of launches continues its exponential growth, and ITU filings for non-GSO satellites to be launched by 2030 have now reached 1.7 million. While attention is shifting towards security & defence, concerns about the safety and sustainability of the space environment must continue to be heard and synergised with emerging priorities. In this environment, ensuring that space remains accessible and usable in the long-term requires communication, coordination and cooperation among European stakeholders. To facilitate such communication, **on 28 March 2025, ESPI held the workshop on "European Cooperation in Space Sustainability".** The workshop was organised in the **framework of the inauguration of ESPI's new Centre for Sustainability and Resilience (CSR).**

The discussion welcomed around 30 attendees hailing from national governments, the European Commission, the European Parliament, the European Space Agency, and industry. The discussion was structured in **three working sessions,** focusing on **identifying key space safety and sustainability subjects, elucidating arguments to advance European space safety ambitions, and proposing next steps for the community.**

As the 1st event of the new ESPI CSR after the Centre's inauguration, the workshop was purposefully broad to elucidate the key topics and subjects for future discussions. ESPI will continue the conversations started during this workshop, and we invite ESPI members and the broader community to propose ideas and topics to further shape the debate in the framework of the Centre.



PANEL I

Identification of Key Space Safety and Sustainability Topics

In the first part of the workshop, participants focused on what they deem the most important topics in space safety and sustainability for Europe, highlighting the need to back up policy and regulatory action with the **development of capabilities.** Key common threads emerged from live polling (see Figure 1) and the conversation: **a focus on Space Situational Awareness (SSA), including topics on debris mitigation and collision risks; in-orbit servicing; as well as the need to develop Space Domain Awareness (SDA) capabilities and infrastructure resilience.**

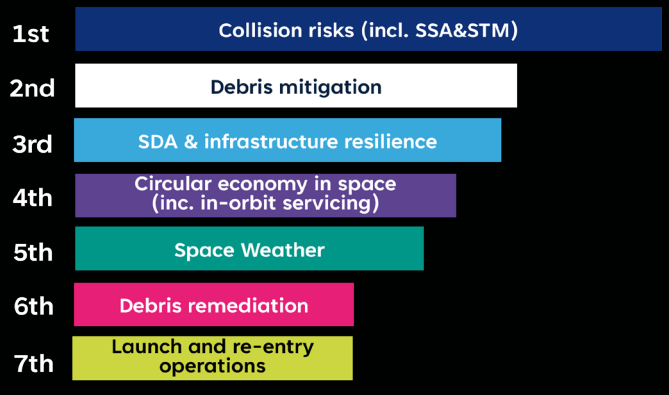


Figure 1: Live polling results - Which areas should the emphasis be put on?

KEY TAKEAWAYS

EUROPEAN COOPERATION IN SPACE SUSTAINABILITY



1 More trust is needed for efficient European SSA

Though Europe has advanced its SSA & SST capabilities in recent years, particularly through the EU SST Partnership, several outstanding issues remain. While the EU SST Partnership allows Europe to pool its resources and strengthen its autonomous SST capabilities, Europe still largely remains dependent on U.S. data-sharing and information centres.

Even though the respective strengths of ESA and the EU are better defined at the European level, **a key topic of discussion was the ongoing potential duplication of efforts by Member States. Such duplication often arises from the lack of trust between Member States** — be it on sharing their SSA cataloguing systems or even plainly stating their SSA capability requirements. The lack of multilateral agreements for SSA data-sharing in Europe, or a comprehensive set of agreements among EU member states, leads to splintered SSA capabilities. While there could be limited reasons to withhold some data based on national security considerations, **increasing trust and sharing more SSA data and information between Member States would vastly improve Europe's joint SSA capabilities.** This would allow countries to channel funding to unique, value-adding capabilities.

2 Harness the opportunities of In-Orbit Servicing



Building on the recent Joint Declaration for the deployment of the In-Space Operations and Services Pilot Mission by five EU Member States (DG DEFIS 2025), **workshop participants highlighted the need to develop a broad range of new solutions for satellite operators based on Rendezvous and Proximity Operations (RPOs) techniques as a key opportunity for Europe in the field of space safety and sustainability.** This will also include streamlining European efforts in areas such as in-orbit servicing. The current approach to the management of space assets was compared to that of a driver who simply leaves their car in the middle of the road the first time it runs out of gas and buys a new one. Finding ways to instead rebuild and repair space assets was discussed as a game-changer that Europe can win the race for to increase its competitiveness and raise funds for further space pursuits.

Notwithstanding, legal and regulatory challenges, including ownership of objects, intellectual property concerns, liability and insurance issues, as well as standardisation challenges for the development of multipurpose, flexible and interoperable solutions still need to be addressed. In addition, developing technologies facilitating proximity operations could also bolster Europe's on-orbit prowess for military actors - highlighting a clear link between "sustainability" and "resilience".

3 Increasing European space capabilities need to back up further regulatory efforts



Several attendees noted that while the costs of regulatory compliance might be high, they appreciate the certainty that cohesive regulatory frameworks provide. Yet, **for European regulatory efforts to become credible, they must be bolstered by a concerted effort to develop European space capabilities.**

Strengthening capabilities, with a particular focus on mitigation and then on remediation measures, will give Europe a stronger negotiating position, and fill the leadership void in current international frameworks, especially in certain areas such as space traffic coordination. Thus, multiple attendees underlined that **strengthening Europe's capabilities remains a main goal for the near future,** and it is the key element to enabling Europe's leadership ambitions.

PANEL II

Prioritisation of Key Elements to Advance European Ambitions

In the second working session, participants discussed the key technological and political arguments for advancing a European ambition in space safety and sustainability. Two common threads emerged from the conversation and live polling: **dangers unsustainable practices pose to citizens and the economy, and the intersections of the topic with security and defence considerations.**

4 Unsustainable space practices directly impact human lives and the economy

One story on debris has been on everyone's mind lately — the uncontrolled re-entry of SpaceX's Falcon 9 rocket raining space junk all over Poland. (BBC 2025) The timing and impact locations were fortuitous, as the debris fell at night and avoided homes, resulting in no casualties. **Yet, as uncontrolled re-entry cases multiply, so, too, will the risk of debris taking human lives.** Anticipating and avoiding such disasters, including through the provision of timely data, instead of addressing them only after they take place, is a key challenge for Europe's (space) safety.



A key consideration for participants has also been the negative impact caused by the lack of efforts or services in space safety with a **diffused effect across the economy.** Unsustainable practices in space are already impacting the Earth, costing people and businesses both time and money. Flights have been diverted and cancelled in the U.S. due to debris from breaking up rockets, leaving thousands of people stranded and demanding compensation (Reuters 2025). A severe Space Weather event in May 2024 disrupted an entire fleet of self-driving tractors that rely on GPS, costing the agricultural industry hundreds of millions of euros (The New York Times 2024). As pre-operational Space Weather service provision in the European context reaches maturity, it also becomes crucial to explore the needs and challenges for the Space Weather community, including the transition pathways to operational services.



Thus, **the biggest immediate threat unsustainable space practices pose to Earth** is not necessarily the potential big "black swan" tragedy, but rather **the diffused economic impacts** that can compound and multiply to billions of euros. **The costs of these impacts, however, can be offset with proper services, capabilities and commitments** focused on space safety and sustainability.

5 Security & Defence is emerging as a pillar in the discussion

In the new geopolitical realities, **security and defence considerations are taking centre-stage and should be effectively used as a guide toward renewed ambitions in Europe.** While the security dimension in the safety and sustainability context has been, for a long time, secondary in Europe, the live polling results (see Figure 2) show a consensus on its high impact, recognising the need to leverage defence and security capabilities for space safety and sustainability goals - and vice versa.

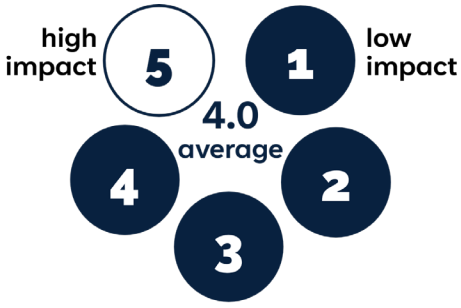


Figure 2: Live polling results – Does the increased focus on security and defence impact space safety and sustainability priorities?

Europe lacks the necessary capabilities to engage in fully autonomous surveillance, tracking, and cataloguing for SDA purposes. Building on this need, the EU, in particular the EDF, is actively funding new projects, many concerning SSA capabilities. Another example is provided by the space surveillance radar developed by Indra (through an agreement with the ESA), recently under contract to equip the German Air Force.

In addition, protecting European space assets from various negative impacts of the space environment — whether they comprise debris from the ASAT tests or Space Weather events — is becoming necessary to ensure their continued operability in times of potential crisis.

A dual framework is essential for effectively addressing emerging safety risks, sustainability challenges, and security and defence threats while aligning approaches and uniting European actors.

PANEL III

From talk to action: Implementing Steps

In the third part of the workshop, participants focused on identifying and evaluating immediate implementing steps concerning space safety and sustainability for Europe, with the discussion and live polling (see Figure 3) largely revolving around ways to improve cooperation within the European space ecosystem. Key common threads emerged from the conversation: **a focus on supporting further capability-building in Europe and the increasing need for leveraging like-minded partners' support.**

6 Focus on capability-building efforts, not streamlining governance

Already possessing most of the building blocks to play a leading role in space safety and sustainability, **European institutions, bodies, and governments should leverage each other's respective strengths and streamline work toward a shared and complementary vision and complementary capabilities.** This effort is particularly timely in view of the upcoming financial milestones, namely CM25 and the next MFF, recognising the need for Europe to continue playing a leading role to ensure its long-term competitive edge.

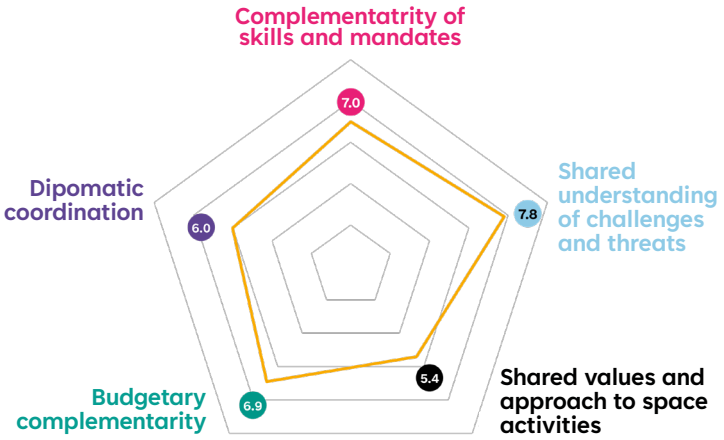


Figure 3: Live polling results – What are the drivers for enhancing cooperation between ESA-EU-MSs?

As debates concerning European space governance structures and approaches will unlikely reach an end, it is time to instead focus on acknowledging each other's competencies, expertise and capacities, and through this shared yet diffused approach, solidify European capabilities.

To improve the interoperability of national systems, and pursue this shared direction, there is also a **recognised need for coordination between the different ministries and departments within EU and ESA Member States.** Moreover, a proposed practical implementing step related to streamlining communication and coordination between ESA and the EU regarding technical requirements for the development of national capabilities under ESA and EU frameworks. A further necessity is discussing the potential standardisation of technical requirements between ESA and the EU, enabling interoperability of the wider European architecture. For example, ESA and Member States could coordinate with the EU SST Partnership at the planning stages for a new national SSA asset to clarify whether it fits into the Partnership's architecture and if the EU will ultimately be able to bear the cost of its operations. This would allow Europe to leverage Member States willing to participate in the EU and ESA programmes or initiatives and enable them to pre-assess the technical specification requirements to avoid the misallocation of their resources.

7 Enhanced collaboration with like-minded partners is key

Both established and emerging spacefaring nations in Europe have an interest in increasing their space capabilities. When pooled together, their individual, national space capabilities and expertise might be considered as a source for high-quality services, at the cutting-edge of the most advanced world standards and functioning within an adequate framework.

The current geopolitical environment is increasingly adversarial, exacerbating risks to space safety and sustainability rather than mitigating them. With the progressive fragmentation of the current geopolitical landscape, participants highlighted the urgent need for effectively leveraging and supporting the mandate of international bodies such as UN COPUOS. This will help coordinate with space powers like the U.S. and China, and solidify Europe's leading position.

Engagement and collaboration with stakeholders outside of Europe represents a key pillar for ensuring that space safety and sustainability remain central to the conversation, fostering coalitions, establishing diplomatic channels, and leveraging European industry. On a bilateral level, the EU and Member States are interested in further pursuing collaboration with like-minded partners, as well as strengthening relations with NATO.

In this context, Europe must be sure to develop better capabilities and achieve greater technical and political autonomy to unlock new opportunities for partnerships, particularly with mid-powers and rising space actors.